

Task 2 study metadata audit: Pattern A → intermediate → Pattern B

Pattern	Study	Species	Protocol, duration & intensity	Post time	Muscle/site	Design	Library & instrument	Sex/training
A	GSE108643	human	Cycling; ~92–100 min to expend 650 kcal; 50% VO2max	post(0h)	vastus lateralis	Paired pre/post subjects	polyA RNA; R; SINGLE; Illumina HiSeq 2500	male; Not reported
A	GSE86931	human	Cycling; 70 min; 70% VO2max	post(4h)	vastus lateralis	Paired pre/post subjects	polyA RNA; Not reported; SINGLE; Illumina HiSeq 2000	male; trained
A	GSE126962	mouse	Treadmill; 60 min; ramp 6→16 m/min, 5% incline	post(4h)	gastrocnemius	Independent exercise/control groups	ribo-zero RNA; RF; PAIRED; Illumina HiSeq 2500	male; Not reported
A	GSE132520	mouse	Treadmill; submaximal ramp 10→23 m/min, 10° incline	post(3h)	quadriceps	Independent exercise/control groups	polyA RNA; RF; PAIRED; Illumina NovaSeq 6000	male; Not reported
Intermediate	GSE151066	human	Cycling; 40 min; 60–70% HRR	post(0h)	vastus lateralis	Paired pre/post subjects	polyA RNA; FR; PAIRED; Illumina NovaSeq 6000	male; female; sedentary; active
B	GSE71972	human	Unilateral knee extension; 60 min; 60% peak power	post(0h)	vastus lateralis	Paired pre/post subjects	polyA RNA; R; SINGLE; Illumina HiSeq 2500	Not reported; active
B	GSE87748	human	Exercise protocol not reported; 15 min; 80% VO2max	post(4h)	vastus lateralis	Paired pre/post subjects	ribo-zero RNA; R; SINGLE; Illumina HiSeq 2500	male; sedentary
B	GSE97718	mouse	Graded treadmill to exhaustion; one session	post(3h)	quadriceps	Independent exercise/control groups	polyA RNA; RF; PAIRED; Illumina HiSeq 2500	male; Not reported